

## Problem Set 8

Solutions by Harsh Bhalala

---

**Problem 1. Strong Connectivity**

(a) In a directed graph  $G$  equivalence relation  $R$  on the vertices of  $G$  as follows:  $u R v$  is true IFF there exists a directed walk from  $u$  to  $v$  and a directed walk from  $v$  to  $u$ . (Note: A walk from  $u$  to  $v$  is expressed as  $u \rightarrow v$ ) Now to show that the given relation is an equivalence relation:

- **Reflexive:** For any vertex  $u$ , there always exists walk of length from  $u$  to  $u$  in both directions, so  $u R u$ .
- **Symmetric :** Suppose  $u R v$ , There then there exists  $(u \rightarrow v)$  and  $(v \rightarrow u)$ , Which are exactly two walks required for relation  $u R v$ .
- **Transitive:** If  $a R b$  and  $b R c$  then we know there exists walks  $(a \rightarrow b)$ ,  $(b \rightarrow a)$ ,  $(b \rightarrow c)$ ,  $(c \rightarrow b)$  and by concatenating those walk we get  $(a \rightarrow b \rightarrow c)$  which implies  $(a \rightarrow c)$  and similarly, by doing the same for the other two walks we get  $(c \rightarrow a)$ . Hence  $b R c$ .

(b) By comparing the definition of strongly connected components and  $R$  we observe that The equivalence classes of  $R$  are exactly the strongly connected components.

**Problem 2. Bijections via Inverses**

(a)

*Proof.* Let  $a_1, a_2 \in A$  such that  $f(a_1) = f(a_2)$ , now applying  $g$  to both side,  $g(f(a_1)) = g(f(a_2))$  since  $\forall a, g(f(a)) = a$ ,  $a_1 = a_2$ . So  $f$  is surjective.  $\square$

(b)

*proof by contradiction.* Assume  $f$  is not surjective. Hence,  $\exists b \in B$  such that  $\forall a \in A, f(a) \neq b$ . But  $f(g(b)) = b$  is true for all  $b \in B$  which is a contradiction since  $g(b)$  is an element of  $A$  that maps to  $b$ .  $\square$

(c) From parts (a) and (b), we know  $f$  is a function, total, injective, and surjective. Hence  $f$  is a bijection.

**Problem 3. Increasing Correspondence**

(a) Let  $(a, b, c) \in A$ . Now by the definition  $f(a, b, c) = (c - b, b - a, a)$  and because  $a \leq b \leq c$  we observe that  $c - b \geq 0$ ,  $b - a \geq 0$  and  $a \geq 0$ , which implies that for every valid input pair there is at least one output. Therefore, the given function is a total function.

Now we show that every output of  $f$  lies in set  $B$ . As we know  $\forall (a, b, c) \in A$ ,  $f(a, b, c) = ((c - b, b - a, a))$ , and given pair is part of set  $B$  IFF

$$(c - b) + 2(b - a) + 3a = a + b + c = 1000$$

which satisfies all necessary conditions and by this we can conclude that  $\forall (a, b, c) \in A$ , so  $f((a, b, c)) \in B$ .

(b) Total function  $g : B \rightarrow A$ ,  $g(x, y, z) := (z, y + z, x + y + z)$   
 $\forall (x, y, z) \in B$ , triples  $(z, y, x)$  is defined over natural number and since addition is deterministic operation  $(z, y + z, x + y + z)$  will garentes exactly one output for all possible input. Also  $x + y + z \geq y + z \geq z$  is true because  $y, x \geq 0$ . Sum:  $z + (y + z) + (x + y + z) = x + 2y + 3z = 1000$ . So  $g(x, y, z) \in A$ .

$$g \circ f = g(f(a, b, c)) = g(c - b, b - a, a) = (a, (b - a) + a, (c - b) + (b - a) + a) = (a, b, c).$$

$$f \circ g = f(g(x, y, z)) = f(z, y + z, x + y + z) = ((x + y + z) - (y + z), (y + z) - z, z) = (x, y, z).$$

(c) Since we proved that  $g$  is a two-sided inverse of  $f$ , and that is necessary and sufficient to prove that  $f$  is bijection

## Problem 4. Counting Graphs

(a)

$$2^{\frac{n(n-1)}{2}}$$

(b)

$$n \times 4 \times 3 \times 3 = 36n$$

## Problem 5. Counting

(a) There are total  $21!$  ways to arrange 21 consonants. Now, since every vowel appears to the left of a consonant, we must choose 5 total consonants to make a gap in front and place a vowel. There are  $\binom{21}{5}$  ways to make such a gap, and now there are  $5!$  ways to put a vowel into those gaps.

Hence total way =

$$21! \times \binom{21}{5} \times 5!$$

(b) We first need to make a total of 5 [VCC] boxes and the remaining 11 consonants in [C] boxes:

$$\frac{\left(\binom{21}{2} \times \binom{19}{2} \times \binom{17}{2} \times \binom{15}{2} \times \binom{13}{2} \times 2^5\right)}{5!} \times 5! \times 16!$$

We needed to divide by  $5!$  because the count is for distinct sections of groups, meaning (bc, df,...) is the same as (df, bc,...) and  $2^5$  because pairs bc and cb are distinct. and there are  $5!$  ways to place vowels in each box. and there are a total of 11 boxes of [C], so the number of ways to arrange 16 distinct boxes is  $16!$

And simplifying the given equation gives:

$$\frac{21! \times 16!}{11!}$$

(c) (Note: Another concise way to express the equation in (b) is used here, which is via the multinomial coefficient) Need to choose  $n$  partitions each of size 2, from  $2n$  choices that can be done in:

$$\binom{2n}{2, 2, 2, \dots, 2} \times \frac{1}{n!} = \frac{(2n)!}{2^n n!}$$

We must here, just like in part (b), the binomial coefficient gives an ordered pair, and since there are a total of  $n$  terms, there are  $n!$  copies of same arrangement.

(d) We can simply apply the Stars and Bars combinatorics theorem. Here we have  $n$  stars and 9 bars, (since there are 10 digits 0 to 9 and each partition says how many times a digit appears in the number) and we

need to find the number of ways to arrange them. The total number of positions to place bars is  $n + 9$  and we need to choose 9 positions for bars, which gives us the following expression:

$$\binom{\text{total position to place bars}}{\text{bars}} = \binom{\text{stars+bars}}{\text{bars}} = \binom{n+9}{9}$$